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Multi-node and Multi-GPU Neural Physics Modelling

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Abstract

Natural phenomena can be modeled using differential equations that describe the physical processes. A numerical solver can be employed by discretizing the differential equations. The solver is inspired by a convolutional neural network strategy on non-uniform structured meshes. Computational fluid dynamics (CFD) is an important modelling of many applications in physics. The worked example is the flow past a sphere. However, CFD is a computer-intensive task. A domain decomposition is applied for the parallel version of the neural physics solver. The parallel processing is prepared to be executed on a heterogeneous architecture combining multi-CPU and multi-GPU. The developed parallel version presents a quasi-linear speed-up.

Keywords

Computational fluid dynamics, convolutional network solver, CPU multi-core, GPU co-processor, High performance heterogeneous architecture

Introduction

Natural phenomena can be well represented by equations relating a balance of quantities on a differential scale. Several applications can be simulated by solving those differential equations. Fluid flow is a good example of a physical process modeled by differential equations. In fact, computational fluid dynamics is a field with many applications, including computational aerodynamics, numerical weather prediction, ocean circulation, hydrology, environmental pollution, the spread of airborne pathogens, and hemodynamics in the body.

Computational Fluid Dynamics: AI4PDE

Computational fluid dynamics (CFD) is a fundamental issue in scientific computing, due to important applications in many areas: engineering, aircraft design, geophysical fluid dynamics (weather, ocean, hydrology), environmental sciences, and human health issues. Here, the applied CFD approach is the *AI4PDE* code (Nadimy et al. 2026(@). It is a code developed by the Applied Modelling and Computation Group (AMCG), Department of Earth Science and Engineering of Imperial College London (UK).

This code solves the Navier-Stokes equations by a convolutional network strategy Fluidity Manual (2025); **?**); **?**); **?**. The selected type of isothermal fluid flow was *flow past a sphere* – see: Campregher et al. (2009); Shang et al. (2021), and Fluidity Manual (2025), page 232. The simulated scenario describes a relevant point in fluid flow development. The worked example can be associated with the *D'Alembert's Paradox* **?**: in an idealized fluid, the viscosity could be neglected, and the drag force over a submerged solid body is zero with this assumption – the *paradox*.

The equations of motion for the incompressible fluid are as follows:

$$\frac{\partial \mathbf{u}(\mathbf{x},t)}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u} - \nabla p = -\frac{1}{R_e}\left[\nabla\cdot\tau + \mathbf{F}\right] \quad (1)$$
$$\text{with:} \quad (\mathbf{x},t) \in \Omega \oplus \mathbb{R}^+$$

where $\mathbf{u}(\mathbf{x},t)$ is the velocity vector: $\mathbf{u} = \{u, v, w\}$, p is the pressure, R_e is the Reynolds number, τ is the stress tensor, and $\mathbf{F}$ is a forcing term. The space domain is defined by $\Omega = \prod_{\alpha=1}^{3}(-L_{\alpha_1}, L_{\alpha_2})$ — where $\alpha = (x, y, z)$, with the symbol $\prod$ indicating the interval Cartesian products. Dirichlet boundary conditions $\partial\Omega$ are given by:

$$\mathbf{u}(\mathbf{x},t)|_{\mathbf{x}\in\partial\Omega} = \begin{cases} g(t)\,, & \text{for: } \mathbf{x} \in [-L_{x_1}, y, z] \\ 0\,, & \text{for: } \mathbf{x} \notin [-L_{x_1}, y, z] \end{cases} . \quad (2)$$

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The pressure field $p(\mathbf{x},t)$ is in balance, according to the Poisson equation ($\mathbf{x} \in \Omega$) and boundary condition ($\mathbf{x} \in \partial\Omega$):

$$\nabla^2 p(\mathbf{x},t) = \nabla \cdot \left\{ \frac{\partial \mathbf{u}(\mathbf{x},t)}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} - \frac{1}{R_e}\left[\nabla \cdot \tau + \mathbf{F}\right] \right\} \quad (3)$$

$$\left.\frac{\partial p}{\partial \mathbf{n}}\right|_{\mathbf{x}\in\partial\Omega} = \frac{\partial \mathbf{u_n}}{\partial t} + \mathbf{u_n} \cdot \nabla \mathbf{u_n} - \frac{1}{R_e}\left[\nabla \cdot \tau + \mathbf{F}\right] \quad (4)$$

with $\mathbf{n}$ being the normal vector to the boundary surface $\partial\Omega$. Equations (1) and (3) are solved by employing AI4PDE strategy – see Section below. For a compressible fluid, the pressure equation is obtained by the time derivative of the thermodynamic gas state equation.

AI4PDE solver

AI4PDE adopts the strategy from the convolution neural network algorithm. The scheme is described here.

(i) Comments;
(ii) comments;
(iii) comments.

Fluid Flow Simulation: AI4PDE

The sagej class file should run on any standard LaTeX 2_ε installation. If any of the fonts, style files or packages it requires are missing from your installation, they can be found on the *TEX Collection* DVDs or downloaded from CTAN.

The article header information

The heading for any file using sagej.cls is shown in Figure 1. You must select options for the trim/text area and the reference style of the journal you are submitting to. The choice of `options` are listed in Table 1.

Table 1. The choice of options.

Option	Trim and font size	Columns
`shortAfour`	210 × 280 mm, 10pt	Double column
`Afour`	210 × 297 mm, 10pt	Double column
`MCfour`	189 × 246 mm, 10pt	Double column
`PCfour`	170 × 242 mm, 10pt	Double column
`Royal`	156 × 234 mm, 10pt	Single column
`Crown`	7.25 × 9.5 in, 10pt	Single column
`Review`	156 × 234 mm, 12pt	Single column

Option	Reference style
`sageh`	SAGE Harvard style (author-year)
`sagev`	SAGE Vancouver style (superscript numbers)
`sageapa`	APA style (author-year)

For example, if your journal is short A4 sized, uses Times fonts and has Harvard style references then you would need

```
\documentclass[ShortAfour,times,sageh]{sagej}
```

Most *SAGE* journals are published using Times fonts but if for any reason you have a problem using Times you can easily resort to Computer Modern fonts by removing the `times` option.

'Review' option

Some journals (for example, *Journal of the Society for Clinical Trials*) require that papers are set single column and with a larger font size to help with the review process. If this is a requirement for the journal that you are submitting to, just add the `Review` option to the `\documenclass[]{sagej}` line.

Remarks

(i) In `\runninghead` use '*et al.*' if there are three or more authors.
(ii) For multiple author papers please note the use of `\affilnum` to link names and affiliations. The corresponding author details need to be included using the `\corrauth` and `\email` commands.
(iii) For submitting a double-spaced manuscript, add `doublespace` as an option to the documentclass line.
(iv) The abstract should be capable of standing by itself, in the absence of the body of the article and of the bibliography. Therefore, it must not contain any reference citations.
(v) Keywords are separated by commas.
(vi) If you are submitting to a *SAGE* journal that requires numbered sections (for example, IJRR), please add the command `\setcounter{secnumdepth}{3}` just above the `\begin{document}` line.

The body of the article

Mathematics

sagej.cls makes the full functionality of $\mathcal{A}\mathcal{M}\mathcal{S}$TEX available. We encourage the use of the `align`, `gather` and `multline` environments for displayed mathematics. amsthm is used for setting theorem-like and proof environments. The usual `\newtheorem` command needs to be used to set up the environments for your particular document.

Figures and tables

sagej.cls includes the graphicx package for handling figures.

Figures are called in as follows:

```
\begin{figure}
\centering
\includegraphics{<figure name>}
\caption{<Figure caption>}
\end{figure}
```

For further details on how to size figures, etc., with the graphicx package see, for example, **?** or Mittelbach and Goossens (2004).

The standard coding for a table is shown in Figure 2.

Cross-referencing

The use of the LaTeX cross-reference system for figures, tables, equations, etc., is encouraged (using `\ref{<name>}` and `\label{<name>}`).

```
\documentclass[<options>]{sagej}

\begin{document}

\runninghead{<Author surnames>}

\title{<Initial capital only>}

\author{<An Author\affilnum{1},
Someone Else\affilnum{2} and
Perhaps Another\affilnum{1}>}

\affiliation{<\affilnum{1}First and third authors' affiliation\\
\affilnum{2}Second author affiliation>}

\corrauth{<Corresponding author's name and full postal address>}

\email{<Corresponding author's email address>}

\begin{abstract}
<Text>
\end{abstract}

\keywords{<List keywords>}

\maketitle

\section{Introduction}
.
.
.
```

Figure 1. Example header text.

```
\begin{table}
\small\sf\centering
\caption{<Table caption.>}
\begin{tabular}{<table alignment>}
\toprule
<column headings>\\
\midrule
<table entries
(separated by & as usual)>\\
<table entries>\\
.
.
.\\
\bottomrule
\end{tabular}
\end{table}
```

Figure 2. Example table layout.

End of paper special sections

Depending on the requirements of the journal that you are submitting to, there are macros defined to typeset various special sections.

The commands available are:

```
\begin{acks}
To typeset an
  "Acknowledgements" section.
\end{acks}

\begin{biog}
To typeset an
  "Author biography" section.
\end{biog}

\begin{biogs}
To typeset an
  "Author Biographies" section.
\end{biogs}

\begin{dci}
To typeset a "Declaration of
  conflicting interests" section.
\end{dci}

\begin{funding}
To typeset a "Funding" section.
\end{funding}

\begin{sm}
To typeset a
  "Supplemental material" section.
\end{sm}
```

Endnotes

Most *SAGE* journals use endnotes rather than footnotes, so any notes should be coded as `\endnote{<Text>}`. Place the command `\theendnotes` just above the Reference section to typeset the endnotes.

To avoid any confusion for papers that use Vancouver style references, footnotes/endnotes should be edited into the text.

References

Please note that the files SageH.bst and SageV.bst are included with the class file for those authors using BIBTEX. The files work in a completely standard way, and you just need to uncomment one of the lines in the below example depending on what style you require:

```
%%Harvard (name/date)
%\bibliographystyle{SageH}
%%Vancouver (numbered)
%\bibliographystyle{SageV}
\bibliography{<YourBibfile.bib>}
```

and remember to add the relevant option to the `\documentclass[]{sagej}` line as listed in Table 1.

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This class file was developed by Sunrise Setting Ltd, Brixham, Devon, UK.
Website: http://www.sunrise-setting.co.uk

Conclusion

In this study, we extend our numerical framework for solving the Shallow Water Equations by introducing non-uniform structured grids. This enables us to have subdomains of different resolutions to capture important hydrodynamic features efficiently. The discretised equations are formulated as discrete convolutions, interpretable as untrained convolutional layers within a neural network. Communication between the subdomains through halo nodes is managed using precomputed indices, allowing efficient, vectorised updates across interfaces. The approach is platform-independent and could be integrated with trained neural networks, such as sub-grid or surrogate models. The results match analytical solutions for a circular dam break, and show good agreement with observations and existing models for the 2005 Carlisle flood. The use of non-uniform grids reduces computational cost, enabling scalable modelling for large domains.

References

Nadimy A, Chen B, Chen Z, H CE and Pain CC (2026): Solving the discretised shallow water equations using non-uniform grids and machine-learning libraries. *Environmental Modelling and Software*, vol. 196, 106752 – DOI: https://doi.org/10.1016/j.envsoft.2025.106752 .

Campregher R, Militzer J, Mansur SS and Silveira Neto A. (2009): Computations of the Flow Past a Still Sphere at Moderate Reynolds Numbers Using an Immersed Boundary Method. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, vol. XXXI, No. 4, 344–352 – DOI: https://doi.org/10.1590/S1678-58782009000400009 .

Shang, W, Zhao H, Li D, Luo K and Fan J (2021): Direct numerical simulation of the flow around a sphere immersed in a flat-plate turbulent boundary layer. *Physics of Fluids*, vol. 33, No. 11, 115106 – DOI: https://doi.org/10.1063/5.0071878 .

Applied Modelling and Computation Group (AMCG) (2025): Fluidity Manual. Imperial College London. https://github.com/FluidityProject/fluidity/tree/main/manual .

Mittelbach F and Goossens M (2004) *The LATEX Companion*, 2nd edn. Addison-Wesley.